7. Integration

What students need to learn:

Indefinite integration as the reverse of differentiation.	Students should know that a constant of integration is required.
Integration of x^n.	For example, the ability to integrate expressions such as $\frac{1}{2}x^2 - 3x^{-\frac{1}{2}}$ and $\frac{(x+2)^2}{x^{\frac{1}{2}}}$ is expected. Given $f'(x)$ and a point on the curve, students should be able to find an equation of the curve in the form $y = f(x)$.
Evaluation of definite integrals.	
Interpretation of the definite integral as the area under a curve.	Students will be expected to be able to evaluate the area of a region bounded by a curve and given straight lines. E.g. find the finite area bounded by the curve $y = 6x - x^2$ and the line $y = 2x$. $\int x \, dy$ will not be required.
Approximation of area under a curve using the trapezium rule.	For example, evaluate $\int_0^1 \sqrt{(2x+1)} \, dx$ using the values of $\sqrt{(2x+1)}$ at $x = 0, 0.25, 0.5, 0.75$ and 1.